GROUP C*-ALGEBRAS: REDUCED, UNIVERSAL, AND EXOTIC

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CERTIFICATE

This is to certify that **Tanvi Telang**, BS-MS (Dual Degree) student in Department of Mathematics, has completed bonafide work on the thesis entitled **'Group C*-algebras: Reduced, Universal, and Exotic'** under my supervision and guidance.

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ABSTRACT

This thesis starts with an introduction to the theory of C*-algebras, the spectral theory of operators and the GNS construction. The aim is to describe the construction of group C*-algebras for a discrete group Γ. We start with two fundamental constructions, the reduced group C*-algebra, $\mathrm{C}^*_r(\Gamma)$, and the universal group C*-algebra, $\mathrm{C}^*(\Gamma)$. We then explain the construction of exotic C*-algebras, $\mathrm{C}^*_D(\Gamma)$ constructed via algebraic ideals $D \lhd \ell^\infty(\Gamma)$. The first main result states that the for the case $D = \ell^p(\Gamma)$, $p \in [1, 2]$, we have $\mathrm{C}^*_D(\Gamma) = \mathrm{C}^*_r(\Gamma)$ and for $D = \ell^\infty(\Gamma)$, we have $\mathrm{C}^*_D(\Gamma) = \mathrm{C}^*(\Gamma)$.The second result provides an example for the case $\Gamma = \mathbb{F}_n$ where this construction is non-trivial, i.e., there exists $1 < p < \infty$ with $\mathrm{C}^*_{\ell^p}(\Gamma) \neq \mathrm{C}^*_r(\Gamma) \neq \mathrm{C}^*(\Gamma)$. The final result states that Γ is amenable if, and only if, $\mathrm{C}^*_r(\Gamma) = \mathrm{C}^*_{\ell^p}(\Gamma) = \mathrm{C}^*(\Gamma)$ for all $1 \leq p \leq \infty$.

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1. INTRODUCTION

Group C*-algebras are an important class of C*-algebras constructed using the space $\mathbb{C}[G]$ for a locally compact group G. They are studied via representations of the group which correspond to representations of the group algebras. We start with the basic spectral theory of C*-algebras which talks about the spectrum of an element and its properties, followed by the continuous functional calculus and the Gelfand–Naimark–Segal construction. We further explain some basic properties of completely positive maps, the highlights being the Stinespring dilation theorem and the Arveson's extension theorem.

We then introduce a particular class of group C*-algebras associated to discrete groups. We begin by describing unitary representations of groups and the correspondence to representations of their associated group algebras. The constructions involves the completion of the pre-C*-algebra, $\mathbb{C}[\Gamma]$ (which is a $*$-algebra itself) with respect to various norms. The most commonly studied group algebras, the reduced and universal group algebras, are explained. We then describe some basic definitions of group properties that are usually studied, such as amenability. We go on to explain the construction of exotic group C*-algebras that were studied by Brown and Guentner in [2]. These are intermediate group algebras that lie between the $\mathrm{C}^*_r(\Gamma)$ and $\mathrm{C}^*(\Gamma)$

and are constructed using algebraic ideals in $\ell^\infty(\Gamma)$. We conclude with some important results obtained about exotic C*-algebras and their properties.

2. PRELIMINARIES

2.1 C*-algebras and Spectral Theory

This section covers some basic definitions and results that are the building blocks of the theory of operator algebras. We start with the definition of a C*-algebra. It is important to mention that most of the results in this section hold true for arbitrary Banach algebras as well but the focus of this thesis will be on C*-algebras.

Definition 2.1 (C*-algebra)**.** An involutive Banach algebra $\mathcal{A}$ is called a C*-*algebra* if the involution of $\mathcal{A}$ satisfies the following condition, known as the C*-*identity*, for each x in $\mathcal{A}$:

$$\|x^*x\| = \|x^*\|\|x\| = \|x\|^2$$

Lemma 2.2. The invertible elements in a C*-algebra, denoted by $G(\mathcal{A})$, form a group that is open in $\mathcal{A}$.

The proof of this lemma comes from the following result [7]: If $x \in \mathcal{A}$ is in the open unit ball with $\|x-1\| < 1$, then $x \in G(\mathcal{A})$ with $x^{-1} = \sum_{n=0}^{\infty}(1-x)^n$.

Example 2.3 (Examples of C*-algebras)**.** The most common examples of

C*-algebras are given below. We will shortly see that the first two examples play extremely important roles in characterizing C*-algebras.

1. *An abelian* C*-algebra *with identity:* $C(X)$ where X is a compact space with the operations:

$$\begin{aligned} \|f\| &= \sup\{|f(\omega)| : \omega \in \Omega\} \\ (f \pm g)(x) &= f(x) \pm g(x); \quad (fg)(x) = f(x)g(x) \\ f^*(x) &= \overline{f(x)} \end{aligned}$$

   $C(X)$ is a C*-algebra since $\|f^*f\| = \sup_{\|x\|\leq 1} \{|f(x)|^2\} = \|f\|^2$.

2. *An abelian* C*-algebra *without identity:* $C_0(X)$ where X is a locally compact space with the same operations as in $C(X)$.

3. *A non-abelian* C*-algebra *with identity:* $\mathbb{B}(\mathcal{H})$, bounded operators on a Hilbert space $\mathcal{H}$, with the operations:.

$$\begin{aligned} \|T\| &= \sup_{\|x\|\leq 1} |T(x)| \\ (T \pm S)(x) &= T(x) \pm S(x); \qquad (TS)(x) = T \circ S(x) \\ T^* &= \text{ adjoint of } T \end{aligned}$$

   We know that $\|T^*T\| = \|T\|^2$.

4. *Group* C*-algebra*s:* Chapter two discusses these in detail. 3

Remark 2.4. For a finite dimensional Hilbert space of dimension n, $\mathbb{B}(\mathcal{H})$ is the set of $n \times n$ matrices $M_n(\mathbb{C})$. We have seen that for studying matrices,

the eigenvalues and eigenvectors play an important role. This concept of eigenvalues is generalised for an arbitrary algebra by defining a spectrum which gives rise to the spectral theory of C*-algebras.

Definition 2.5 (Spectrum of an element)**.** The *spectrum* of $a \in \mathcal{A}$ is the set

$$\sigma_{\mathcal{A}}(a) = \{\lambda \in \mathbb{C} : (a - \lambda) \text{ is not invertible}\}.$$

The complement $\mathbb{C} \setminus \sigma_{\mathcal{A}}(a)$ is known as the *resolvent* of a.

Definition 2.6 (Spectral radius)**.** For an element $a \in \mathcal{A}$, the spectral radius of a is given by

$$\|a\|_{sp} = \sup\{|\lambda| : \lambda \in \sigma(a)\}.$$

Proposition 2.7. The spectrum of a Banach algebra is a compact subset of $\mathbb{C}$ with $\|x\|_{sp} \leq \|x\|$ for each x. Moreover, the spectral radius can also be given by the formula

$$\|x\|_{sp} = \lim_{n\to\infty} \|x^n\|^{1/n}.$$

Definition 2.8. $\Omega(\mathcal{A})$ is defined to be the set of all non-zero multiplicative linear functionals $\omega : \mathcal{A} \longrightarrow \mathbb{C}$. The set of all maximal ideals in $\mathcal{A}$ is denoted by $\mathcal{M}(\mathcal{A})$.

Proposition 2.9. Any Banach algebra that is a field is isomorphic to $\mathbb{C}$

Proposition 2.10. For an abelian Banach algebra $\mathcal{A}$, there is a bijection between $\Omega(\mathcal{A})$ and $\mathcal{M}(\mathcal{A})$.

Sketch of proof. Given a maximal ideal $\mathcal{M}$, $\mathcal{A}/\mathcal{M}$ is a Banach algebra that is a field, hence it is isomorphic to $\mathbb{C}$. It can be verified that this isomorphic

belongs to $\Omega(\mathcal{A})$. Conversely, given $\omega \in \Omega(\mathcal{A})$, the kernel of the map is a maximal ideal. □

Theorem 2.11 (The Gelfand transform)**.** *The Gelfand transform of an abelian, unital* C*-algebra *is the isometric isomorphism:*

$$\mathfrak{F} : \mathcal{A} \to C(\Omega(\mathcal{A})) \quad a \mapsto \hat{a}$$

where

$$\hat{a} : \Omega(\mathcal{A}) \longrightarrow \mathbb{C} \quad \hat{a}(\omega) = \omega(a).$$

Moreover, $\sigma_{\mathcal{A}}(x) = \hat{a}(\Omega(\mathcal{A}))$*, and* $\|\hat{a}\|_\infty = \|a\|_{sp}$.

Conversely, we also have a result for the case where we initially have a locally compact space Ω.

Proposition 2.12. Let Ω be a locally compact space and $\mathcal{A} = (C_\infty(\Omega))$. The map:

$$\Omega \longrightarrow \Omega(\mathcal{A}) \quad \omega \mapsto \hat{\omega}$$

where $\omega \in \Omega$, and $\hat{\omega}(f) = f(\omega)$ is a homeomorphism of Ω onto $\Omega(\mathcal{A})$.

Proposition 2.13. Let A be a unital C*-algebra. If $x \in \mathcal{A}$ is normal, then there exists a unique isomorphism ϕ of $C(\sigma_{\mathcal{A}}(x))$ onto the C*-subalgebra B of $\mathcal{A}$ generated by x and 1 such that $\phi(1) = 1$ and $\phi(\mathrm{id}) = x$ where id denotes the function $\mathrm{id}(\lambda) = \lambda$ for all $\lambda \in \sigma_{\mathcal{A}}(x)$.

Sketch of proof. Since x is normal, B is abelian. Moreover, it can be seen that the map $\psi : \Omega \longrightarrow \sigma_{\mathcal{A}}(x)$ is a homeomorphism, giving us the map $\phi(f) = \mathcal{F}^{-1}(f \circ \psi)$. □

The image of a continuous function on $\sigma_{\mathcal{A}}(x)$ given by $\phi(f)$ is denoted by $f(x)$ which is an element of $\mathcal{A}$. This is what we call the continuous functional calculus that helps us assign elements of an algebra to continuous functions on the spectrum of an element.

Theorem 2.14 (Spectral Mapping Theorem)**.** *Let x be a fixed element of a unital Banach algebra $\mathcal{A}$. If f is a holomorphic function on a neighborhood of $\sigma_{\mathcal{A}}(x)$, then we have*

$$f(\sigma_{\mathcal{A}}(x)) = \sigma_{\mathcal{A}}(f(x))$$

Furthermore, if g is a holomorphic function on a neighborhood of $f(\sigma_{\mathcal{A}}(x))$, then $f \circ g(x) = g \circ f(x)$

2.2 The Gelfand–Naimark–Segal (GNS) Construction

We have seen that abelian C*-algebras are characterised as spaces of continuous functions on a compact space, using the Gelfand transform. In this section, we explain, using the GNS construction, how a C*-algebra $\mathcal{A}$ can be represented as bounded functions on a Hilbert space $\mathcal{H}$. This is done using *positive linear functionals* on $\mathcal{A}$. ($\mathcal{A} \hookrightarrow \mathbb{B}(\mathcal{H})$).

Let $\mathcal{A}_h$ denote the set of self adjoint elements in $\mathcal{A}$. The set of positive elements $\mathcal{A}_+ \subseteq \mathcal{A}_h$ is characterized by the following theorem.

Theorem 2.15. *For a self-adjoint element x of a* C*-algebra $\mathcal{A}$, the following*

three conditions are equivalent:

1. $\sigma_{\mathcal{A}}(x) \subseteq [0, \infty)$.

2. $x = y^*y$ *for some* $y \in \mathcal{A}$.

3. $x = h^2$ *for some self-adjoint* h.

Moreover, the set P *of all elements* $x \in A_h$ *satisfying any of the above conditions is a closed convex cone in* $\mathcal{A}$ *with* $P \cap (-P) = \{0\}$.

Definition 2.16 (Representation)**.** A representation of $\mathcal{A}$ into $B(\mathcal{H})$, denoted by $(\pi, \mathcal{H})$, is a $*$-homomorphism where $\pi : \mathcal{A} \longrightarrow B(\mathcal{H})$

Definition 2.17 (Positive linear functionals)**.** A linear functional ω on an $\mathcal{A}$ is called *positive* if $\omega(x^*x) \geq 0$ for every $x \in \mathcal{A}$. If $\omega(x^*x) \neq 0$ for every nonzero $x \in \mathcal{A}$, then ω is called *faithful*.

It can be shown that a positve linear functional is continuous.

Theorem 2.18 (Gelfand–Naimark–Segal)**.** *To any positive linear functional* ω *on* $\mathcal{A}$*, there corresponds uniquely, within unitary equivalence, a representation* $\pi_\omega : \mathcal{A} \longrightarrow B(\mathcal{H}_\omega)$ *and* ζ_ω *such that*

1. $\overline{\pi_\omega(\mathcal{A})\zeta_\omega} = \mathcal{H}_\omega$,

2. $\omega(x) = (\pi_\omega(x)\zeta_\omega \mid \zeta_\omega)$, $x \in \mathcal{A}$.

Sketch of proof. [7] For a given ω, we need a Hilbert space, $\mathcal{H}_\omega$, a representation, π_ω and a vector, ζ_ω.

Let $N_\omega := \{x \in \mathcal{A} : \omega(xx^*) = 0\}$. It is a left ideal of $\mathcal{A}$. Now define $\mathcal{H}_\omega = \overline{\mathcal{A}/N_\omega}$. The inner product is given by $\langle x + N_\omega, y + N\omega \rangle = \omega(y^*x)$. The

representation π_ω is given by $\pi_\omega(a)(x + N_\omega) = ax + N_\omega$. Then, $|\langle \pi_\omega(a)x + N_\omega, y + N\omega\rangle| = |\omega(y^*ax)| \leq \|a\|\|x + N\omega\|\|y + N_\omega\|$ giving us that π_ω is continuous. We can extend it to the entire Hilbert space using the continuity of π_ω and the inner product. Now, note that there exists, for $x \in \mathcal{A}$, a unique vector $\zeta_\omega = 1+N_\omega$ such that $\omega(x) = \langle x+N_\omega, \zeta_\omega\rangle$. Hence, $\langle x+N_\omega, y+N\omega\rangle = \omega(y^*x) = \langle y^*x+N_\omega, \zeta_\omega\rangle = \langle x+N_\omega, \pi_\omega(y)\zeta_\omega\rangle$ giving $x+N_\omega = \pi_\omega(y)\zeta_\omega$. Lastly, $\omega(x) = \langle x + N_\omega, \zeta_\omega\rangle = \langle \pi_\omega(y)\zeta_\omega, \zeta_\omega\rangle$ We thus have π_ω to be a bounded linear operator on $\mathcal{H}_\omega$ and the map

$$\pi_\omega : \mathcal{A} \longrightarrow B(\mathcal{H}_\omega), \quad a \mapsto \pi_\omega(a)$$

to be a $*$-homomorphism.

Given two such representations $(\pi_\omega, \mathcal{H}_\omega, \xi_\omega)$ and $\pi'_\omega, \mathcal{H}'_\omega, \xi'_\omega)$ associated to a positive functional ω, define

$$U_0 : \pi'_\omega(\mathcal{A})\xi'_\omega \longrightarrow \pi_\omega(\mathcal{A})\xi_\omega, \quad U_0 : \pi'_\omega(x)\xi'_\omega \mapsto U_0 : \pi_\omega(x)\xi_\omega$$

It can be seen that the extension of U_0 to U is an isometry of $\mathcal{H}'_\omega$ onto $\mathcal{H}_\omega$. It can also be computed to see that for all $x \in \mathcal{A}$, $\pi_\omega(x)U = U\pi'_\omega(x)$.

□

A C*-algebra admits sufficiently many positive linear functionals as we can see from the following result.

Proposition 2.19. In a C*-algebra $\mathcal{A}$, there exists a positive linear functional corresponding to each $a \in \mathcal{A}$ such that $f_a(a^*a) > 0$.

Proof. Let $a \in \mathcal{A}$, $a \neq 0$. Then $-a^*a \notin \mathcal{A}_+$. $\mathcal{A}_+$ is a closed convex cone

in the real Banach Space $\mathcal{A}_h$. By the Hahn–Banach separation theorem, there exists a real linear functional f_a such that $f_a(y) \geq 0 \ \forall y \in \mathcal{A}_+$ and $f_a(-a^*a) < 0$. We can extend it to $\mathcal{A}$ by: $f_a(x+iy) = f_a(x) + if_a(y)$ for all $x, y \in \mathcal{A}_h$. □

Definition 2.20 (Cyclic vector)**.** If a representation $(\pi, \mathcal{H})$ of $\mathcal{A}$ admits a vector ζ such that $[\pi(\mathcal{A})\zeta] = \mathcal{H}$, then $(\pi, \mathcal{H})$ is called cyclic and ζ is called a cyclic vector for π. Here, $[\pi(\mathcal{A})\zeta]$ denotes the closed space generated by $\{\pi(a)\zeta : a \in \mathcal{A}\}$.

Theorem 2.21. *A* C*-algebra *admits a faithful representation. Hence it is isometrically isomorphic to a uniformly closed self adjoint algebra of operators on a Hilbert space.*

Proof. Let $(\pi_a, \mathcal{H}_a, \zeta_a)$ be the cyclic representation of $\mathcal{A}$ induced by f_a where $a \in \mathcal{A}$. $\|\pi_a\zeta_a\|^2 = f_a(a^*a) > 0$ hence $f_a(a) \neq 0$. We get a faithful representation by setting $(\pi, \mathcal{H}) = \sum_{a\in\mathcal{A}\setminus 0}^{\oplus}(\pi_a, \mathcal{H}_a)$. □

2.3 Completely Positive Maps

We have seen earlier that matrices are operators finite dimensional vector spaces. Matrices are also used to study operators on infinite-dimensional spaces using $n \times n$ matrices with entries coming from a C*-algebra $\mathcal{A}$.

Definition 2.22. $M_n(\mathcal{A})$ is the set of $n \times n$ matrices where the entries come from the C*-algebra $\mathcal{A}$.

Definition 2.23. Given two C^*-algebras $\mathcal{A}$ and $\mathcal{B}$ and a map $\phi : \mathcal{A} \to \mathcal{B}$,

we also obtain a map for each $n = 2, 3, \cdots$, given via the formula

$$M_n(\mathcal{A}) \to M_n(\mathcal{B}), \quad \phi_n(a_{i,j} = (\phi\,(a_{i,j}))\,.$$

ϕ is called *completely positive* if every ϕ_n is a positive map. If ϕ is a bounded map, then each ϕ_n is bounded, and when $\|\phi\|_{cb}{:=}\sup_n\|\phi_n\|$ is finite, we call ϕ to be a *completely bounded* map.

We give below a generalization of the GNS construction that helps characterize completely positive maps from a C*-algebra into $\mathbb{B}(\mathcal{H})$.

Theorem 2.24 (Stinespring's Dilation Theorem)**.** *Let $\mathcal{A}$ be a unital* C**-algebra, and let $\phi : \mathcal{A} \to \mathcal{B}(\mathcal{H})$ be a completely positive map. Then there exists a Hilbert space $\mathcal{K}$, a unital $*$-homomorphism $\pi : \mathcal{A} \to B(\mathcal{K})$, and a bounded operator $V : \mathcal{H} \to \mathcal{K}$ with $\|\phi(1)\| = \|V\|^2$ such that*

$$\phi(a) = V^*\pi(a)V$$

Sketch of proof. The proof of this uses the concept of tensor products along with the similar idea used in the GNS construction. We first consider the algebraic tensor product space $\mathcal{A}\otimes\mathcal{H}$ and define the following positive semidefinite bilinear map on it using the inner product on $\mathcal{H}$, $\langle\cdot,\cdot\rangle_{\mathcal{H}}$:

$$\langle a \otimes x, b \otimes x\rangle_{\mathcal{A}\otimes\mathcal{H}} := \langle\phi(b^*a)x, y\rangle_{\mathcal{H}}$$

where $a, b \in \mathcal{A}$ and $x, y \in \mathcal{H}$. The positive semidefiniteness can be checked using the fact that ϕ is completely positive and using the inner product

on $\mathcal{H}^{(n)}$ that we defined earlier. Thus it also satisfies the Cauchy–Schwarz inequality. We define the kernel of this map which is a subspace of $\mathcal{A} \otimes \mathcal{H}$ as

$$\mathcal{N} = \{u \in \mathcal{A} \otimes \mathcal{H} : \langle u, u \rangle = 0\} = \{u \in \mathcal{A} \otimes \mathcal{H} : \langle u, v \rangle = 0 \text{ for all } v \in \mathcal{A} \otimes \mathcal{H}\}.$$

Now we let $\mathcal{K}$ be the completion of $\mathcal{A} \otimes \mathcal{H}/\mathcal{N}$ in the norm obtained by the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$: . Next we need to define π and V. We define

$$\pi(a) : \mathcal{A} \otimes \mathcal{H} \longrightarrow \mathcal{A} \otimes \mathcal{H}$$

$$\pi(a)(\sum_i a_i \otimes x_i) = \sum_i (aa_i) \otimes x_i$$

It can be checked using the Cauchy–Schwarz inequality that $\mathcal{N}$ is invariant under $\pi(a)$, and $\|\pi(a)\| \leq \|a\|$ giving us a well-defined bounded linear operator on $\mathcal{K}$ given by $\pi : \mathcal{A} \to \mathbb{B}(\mathcal{K})$ which is a unital $*$-homomorphism.

Now define

$$V : \mathcal{H} \longrightarrow \mathcal{K}, \quad V(x) = (1 \otimes x) + N$$

Note that $\|V\|^2 = sup\{\langle \phi(1)x, x \rangle_{\mathcal{H}} : \|x\| \leq 1\} = \|\phi(1)\|$. Hence V is bounded, and if ϕ is unital, then V is an isometry. Lastly,

$$\langle V^* \pi(a) V x, y \rangle_{\mathcal{H}} = \langle \pi(a) 1 \otimes x, 1 \otimes y \rangle_{\mathcal{K}} = \langle \phi(a)x, y \rangle_{\mathcal{H}}$$

for all x and y. And hence $V^* \pi(a) V = \phi(a)$. □

When the range C*-algebra is commutative, the concepts of bounded and completely bounded coincide, as do positive and completely positive. A commutative domain is also enough to ensure that positive maps are completely

positive but not completely bounded.

Theorem 2.25 (Arveson's Extension Theorem)**.** *Let $\mathcal{A}$ be a C^*-algebra, $\mathcal{B}$ be a subalgebra in $\mathcal{A}$, and $\phi : \mathcal{B} \to B(\mathcal{H})$ a completely positive map. Then there exists a completely positive map $\psi : \mathcal{A} \to B(\mathcal{H})$ that extends ϕ.*

3. DISCRETE GROUP ALGEBRAS

3.1 Unitary Representations and Group Algebras

Another common occurence of C*-algebras is in the form of group algebras of locally compact groups. Here, we will only consider discrete groups.

Definition 3.1. A discrete group is a topological group equipped with the discrete topology.

We denote a discrete group by Γ and assume that it is countable.

Example 3.2. (Examples of discrete groups)

1. The set of integrs, $\mathbb{Z}$.
2. $SL(n, \mathbb{Z})$, for $n \in \mathbb{N}$ and $n \geq 2$.
3. Free group of order n, $\mathbb{F}_n$.

Definition 3.3. For $1 \leq p \leq \infty$,

$$\ell^p(\Gamma) = \{f : \Gamma \longrightarrow \mathbb{C} : \sum_{s \in \Gamma} |f(s)|^p < \infty\}.$$

$$\ell^\infty(\Gamma) = \{f : \Gamma \longrightarrow \mathbb{C} : \sup_{s\in\Gamma} |f(s)| < \infty\}.$$

Note that $\ell^2(\Gamma)$ is the only Hilbert space.

The group ring and $\ell^1(\Gamma)$*:* The group ring $\mathbb{C}[\Gamma]$ is the set of finite sums $\{\sum_{\substack{s\in\Gamma\\ \text{finite}}} \alpha_n s\}$. The function $s \mapsto \alpha_s$ is an ℓ^1-function if $\sum_{s\in\Gamma} |\alpha_s| < \infty$. Thus, the group ring $\mathbb{C}[\Gamma]$ comes equipped with the ℓ^1-norm $\|\sum_{s\in\Gamma} \alpha_n s\|_1 = \sum_{s\in\Gamma} |a_s|$. In fact, $\overline{\mathbb{C}[\Gamma]}^{\|\cdot\|_1} = \ell^1(\Gamma)$. Γ can be endowed with a $*$-operation given by $s^* = s^{-1}$, $s \in \Gamma$. The space $\mathbb{C}[\Gamma]$ with convolution as multiplication and the involution $\sum_{s\in\Gamma} \alpha_s s = \sum_{s\in\Gamma} \overline{\alpha_{s^{-1}}} s^{-1}$ is a $*$-algebra.

Definition 3.4. For a Hilbert space $\mathcal{H}$, we define a unitary representation of a group

$$\pi : \Gamma \longrightarrow \mathcal{U}(\mathcal{H})$$

to be a group homomorphism which preserves the $*$-operation for $s \in \Gamma$.

Example 3.5. The regular representations are one of the most common and important examples. Here, $\mathcal{H} = \ell^2(\Gamma)$ with the canonical orthonormal basis $\{\delta_t : t \in \Gamma\}$

$$\pi : \Gamma \longrightarrow \mathbb{B}(\ell^2(\Gamma))$$

The left-regular representation is given by $\lambda(s)(\delta_t) = \delta_{st}$. The right-regular representation is given by $\rho(s)(\delta_t) = \delta_{ts^{-1}}$.

In particular, for $\Gamma = \mathbb{Z}$, $(a_n)_{n\in\mathbb{Z}} \in \ell^2(\mathbb{Z})$,

$$\lambda(s)(a_n) = (a_{s+n}) \quad \text{and} \quad \rho(s)(a_n) = (a_{n-s}) \quad s \in \mathbb{Z}.$$

Note that $\rho(s)\lambda(t) = \lambda(t)\rho(s)$ for all $s, t \in \Gamma$:

$$\rho(s)\lambda(t)\delta_e = \rho(s)\delta_{te} = \delta_{tes^{-1}},$$

$$\lambda(t)\rho(s)\delta_e = \lambda(t)\delta_{es^{-1}} = \delta_{tes^{-1}}.$$

Note that, for $f, g \in C_c(\Gamma)$ we have $\lambda(f)g = f * g$. Here we consider $f \in C_c(\Gamma)$ in the domain of λ and $g \in C_c(\Gamma) \subseteq \ell^2(\Gamma)$.

Lemma 3.6. There exists a one-to-one correspondence between unitary representations of a group and representations of the group ring

$$\{\Gamma \to \mathcal{U}(\mathcal{H})\} \longleftrightarrow \{\mathbb{C}[\Gamma] \to \mathbb{B}(\mathcal{H})\}$$

Sketch of proof. Given

$$\pi : \Gamma \longrightarrow \mathcal{U}(\mathcal{H})$$

$$s \mapsto \pi_s$$

we can extend it to

$$\pi : \mathbb{C}[\Gamma] \longrightarrow \mathbb{B}(\mathcal{H})$$

$$\sum_{s\in\Gamma} \alpha_s s \mapsto \sum_{s\in\Gamma} \alpha_s \pi_s$$

$$\|\pi(\sum_{s\in\Gamma} \alpha_s s)\| \leq \sum_{s\in\Gamma} |a_s| = \|\sum_{s\in\Gamma} \alpha_s s\|_1$$

Conversely, given

$$\pi : \mathbb{C}[\Gamma] \longrightarrow \mathcal{U}(\mathcal{H}),$$

the restriction to $\Gamma \subseteq \mathbb{C}[\Gamma]$ gives us a unitary representation of Γ as

$$\pi(s)\pi(s)^* = \pi(s)^*\pi(s) = \pi(1) = \mathrm{id}_{\mathbb{B}(\mathcal{H})}.$$

□

3.2 Reduced and Universal C*-algebras:

The group ring $\mathbb{C}[\Gamma]$ is a pre-C*-algebra. It gives rise to various C*-algebras after completing the space using various C* norms.

We abuse notation and use λ to denote the extension of the left-regular representation to $\mathbb{C}[\Gamma]$.

Definition 3.7 (Reduced C*-algebra)**.** Let $\lambda : \mathbb{C}[\Gamma] \longrightarrow \mathbb{B}(\ell^2(\Gamma))$ be the left-regular representation. For $x \in \mathbb{C}[\Gamma]$, define

$$\|x\|_r := \|\lambda(x)\|_{\mathbb{B}(\ell^2(\Gamma))}.$$

Then, the reduced C*-algebra, $\mathrm{C}^*_r(\Gamma)$ is defined as the completion $\overline{\mathbb{C}[\Gamma]}^{\|\cdot\|_r}$.

Definition 3.8. For $x \in \mathbb{C}[\Gamma]$,

$$\|x\|_u := \sup_{\pi} \|\pi(x)\|_{\mathbb{B}(\mathcal{H})}$$

where π varies over all unitary $*$-representations of Γ. Then the universal C*-algebra, $\mathrm{C}^*(\Gamma)$ is defined as the completion $\overline{\mathbb{C}[\Gamma]}^{\|\cdot\|_u}$. The supremum is well-defined since $\|x\|_u := \sup_\pi \|\pi(x)\|_{\mathbb{B}(\mathcal{H})} \leq \|x\|_1$ for all $x \in \mathbb{C}[\Gamma]$.

Theorem 3.9. *There exists a canonical $*$-homomorphism*

$$\rho : \mathrm{C}^*(\Gamma) \longrightarrow \mathrm{C}_r^*(\Gamma)$$

which extends the identity map on the group algebra $\mathbb{C}[\Gamma]$.

We now give an example of a tracial vector state on $\mathrm{C}_r^*(\Gamma) \subseteq \mathbb{B}(\ell^2(\Gamma))$

Proposition 3.10. Define

$$\omega : \mathrm{C}_r^*(\Gamma) \longrightarrow \mathbb{C}, \quad \omega(x) = \langle x\delta_e, \delta_e \rangle$$

where e is the identity of Γ and $x \in \mathrm{C}_r^*(\Gamma) \subseteq \mathbb{B}(\ell^2(\Gamma))$. Then, ω is a tracial vector state on $\mathrm{C}_r^*(\Gamma)$.

Proof. For any $x \in \mathrm{C}_r^*(\Gamma)$, there exists a sequnece $\{x_i\} \subseteq \mathbb{C}[\Gamma]$ with $x_i \to x$ in the reduced norm. To show that ω is tracial it is enough to show that $\omega(x_i y_i) = \omega(y_i x_i)$ for $x_i = \sum_s \alpha_s \lambda(s)$, $y_i = \sum_t \alpha_t \lambda(t)$ in $\mathbb{C}[\Gamma]$. Note that

$$(\sum_s \alpha_s \lambda(s))(\sum_t \alpha_t \lambda(t)) = \sum_{s,t} \alpha_s \alpha_t \lambda(s)\lambda(t).$$

Since

$$\omega(\lambda_s \lambda_t) = \langle \lambda_s \lambda_t \delta_e, \delta_e \rangle = \langle \lambda_t \lambda_s \delta_e, \delta_e \rangle = \begin{cases} 1, & \text{if } s = t^{-1} \\ 0, & \text{otherwise} \end{cases},$$

we have by linearity that ω is tracial for all $x, y \in \mathrm{C}_r^*(\Gamma)$.

We need to show ω is faithful now. We know (from Example 3.5) that $\rho(s)x = x\rho(s)$ for all $x \in \mathrm{C}_r^*(\Gamma)$. δ_e is a separating vector since if $x\delta_e = y\delta_e$,

then for all $s \in \Gamma$,

$$x\delta_s = \rho_s^* x\delta_e = \rho_s^* y\delta_e = y\delta_s.$$

This implies that $x = y$ since x and y agree on all basis vectors of $\ell^2(\Gamma)$. Now, if $x \geq 0$, $x \in \mathrm{C}_r^*(\Gamma)$ and $\omega(x) = \langle x\delta_e, \delta_e \rangle = 0$, then $\|x^{1/2}\delta_e\| = 0 \implies x = 0$. □

3.3 Some Properties of Γ

Definition 3.11 (Positive-definite functions)**.** $h : \Gamma \to \mathbb{C}$ is positive definite if for every $s_1, \cdots, s_n \in \Gamma$, the matrix $[h(s_i s_j^{-1})]_{i,j} \in M_n(\mathbb{C})$ is positive (semidefinite).

Now, we define the Schur multiplier associated to a function on Γ.

Definition 3.12. Let $\varphi : \Gamma \to \mathbb{C}$ be a function. The associated multiplier is defined by

$$m_\varphi : \mathbb{C}[\Gamma] \to \mathbb{C}[\Gamma], \quad m_\varphi \left(\sum_{t \in \Gamma} \alpha_t \cdot t \right) := \sum_{t \in \Gamma} \varphi(t)\alpha_t \cdot t$$

We also define $\widetilde{m}_\varphi : \lambda(\mathbb{C}[\Gamma]) \to \lambda(\mathbb{C}[\Gamma])$ by

$$\widetilde{m}_\varphi \left(\lambda \left(\sum_{t \in \Gamma} \alpha_t \cdot t \right) \right) = \tilde{m}_\varphi \left(\sum_{t \in \Gamma} \alpha_t \lambda_t \right) = \sum_{t \in \Gamma} \varphi(t)\alpha_t \lambda_t$$

Lemma 3.13. Suppose φ is finitely supported, positive definite, and $\varphi(e) = 1$. Then m_φ extends to a continuous map on $C^*(\Gamma)$ and $\widetilde{m}_\varphi$ extends to a continuous map on $C_r^*(\Gamma)$ both with norm one.

The left-translation action : For $s, t \in \Gamma, f \in \ell^\infty(\Gamma)$, we define an action of Γ on the algebra $\ell^\infty(\Gamma)$ as follows,

$$\Gamma \times \ell^\infty(\Gamma) \longrightarrow \ell^\infty(\Gamma)$$

$$(s.f)(t) = f(s^{-1}t).$$

Definition 3.14 (Amenable group)**.** A discrete group Γ is called *amenable* if there exists a state μ on $\ell^\infty(\Gamma)$ which is invariant under the left-translation action of the group on itself.

Example 3.15. A finite group $S = \{s_i\}_{i=1}^n$ is amenable since the normalised counting measure is a state on $\ell^\infty(S)$ which is invariant under the left-translation action.

We take an interlude and quickly define some notions pertaining to finite dimensional approximations of C*-algebras.

Definition 3.16 (Nuclear map)**.** A map $\theta : \mathcal{A} \to \mathcal{B}$ between C*-algebras is called nuclear if there exist completely contractive positve maps $\phi_n : \mathcal{A} \to M_{k(n)}(\mathbb{C})$ and $\psi_n : M_{k(n)}(\mathbb{C}) \to \mathcal{B}$, where $k(n)$ is a natural number depending on n, such that for all $a \in \mathcal{A}$,

$$\|\psi_n \circ \phi_n(a) - \theta(a)\| \longrightarrow 0.$$

Definition 3.17 (Nuclear C*-algebra)**.** A C*-algebra $\mathcal{A}$ is nuclear if the map

$$\mathrm{id}_{\mathcal{A}} : \mathcal{A} \longrightarrow \mathcal{A}$$

is nuclear.

Definition 3.18 (Exact C*-algebra)**.** A C*-algebra $\mathcal{A}$ is nuclear if there exists a faithful representation π of $\mathcal{A}$ on a Hilbert space $\mathcal{H}$, $\pi : \mathcal{A} \to \mathbb{B}(\mathcal{H})$, such that it is nuclear.

We now state a theorem [3] (without proof) that compiles various characterizations of amenability and also gives a very deep result on the relation between amenability of a discrete group and nuclearity of its corresponding C*-algebras.

Theorem 3.19. *[3, Theorem 2.6.8] Let Γ be a discrete group. The following are equivalent:*

1. *Γ is amenable.*
2. *The trivial representation τ_0 is weakly contained in the regular representation λ (i.e., there exist unit vectors $\xi_i \in \ell^2(\Gamma)$ such that $\|\lambda_s(\xi_i) - \xi_i\| \to 0$ for all $s \in \Gamma$.*
3. *There exists a net (φ_i) of finitely supported positive definite functions on Γ such that $\varphi_i \to 1$ pointwise.*
4. *$C^*(\Gamma) = C^*_\lambda(\Gamma)$.*
5. *$C^*_\lambda(\Gamma)$ is nuclear.*

Although amenability also has multiple other characterizations, we will only need the equivalence of the above given statements.

4. EXOTIC GROUP $\mathbb{C}^*$-ALGEBRAS

4.1 Group $\mathbb{C}^*$-algebras Associated to Algebraic Ideals

We have studied two important C*-algebras associated to any discrete group Γ, the reduced C*-algebra and the universal C*-algebra. In the paper [2] Brown and Guentner describe a class of intermediate group-C*-algebras associated to algebraic ideals of $\ell^\infty(\Gamma)$ other than the most commonly known reduced and universal C*-algebras. They also show that in some cases, for example when Γ is amenable, all these C*-algebras coincide. They also give an example of the free group $\mathbb{F}_n$ that shows that this construction is non-trivial. In fact, in [6], the author shows that for the free group $\mathbb{F}_n$, these group C*-algebras never coincide for a particular special class of ideals, giving rise to a continuum of group C*-algebras.

Definition 4.1. For a unitary representation $\pi : \Gamma \to \mathbb{B}(\mathcal{H})$, the functions

$$\pi_{\xi,\eta} : \Gamma \longrightarrow \mathbb{C} \quad \pi_{\xi,\eta}(s) = \langle \pi(s)\xi, \eta \rangle,$$

where $\xi, \eta \in \mathcal{H}$, are called the matrix coefficients of π. Clearly $\pi_{\xi,\eta} \in \ell^\infty(\Gamma)$.

Given an algebraic ideal D of $\ell^\infty(\Gamma)$ such that $C_c(\Gamma) \subset D$, we define D-represenations and using them, a norm on $\mathbb{C}(\Gamma)$. For example, $\ell^p(\Gamma) = D$ is an alegbraic ideal in $\ell^\infty(\Gamma)$.

Definition 4.2. Let $\pi : \Gamma \to \mathbb{B}(\mathcal{H})$ be a unitary representation. We say that π is a D-representation if there exists a dense subspace $\mathcal{H}_\pi$ of $\mathcal{H}$ such that $\pi_{\xi,\eta} \in D$ for all $\xi, \eta \in \mathcal{H}_\pi$. If π is a D-representation, then the D-norm is given by:

$$\|x\|_D := \sup_\pi \{\|\pi(x)\|_{\mathbb{B}(\mathcal{H})} : \pi \text{ is a } D\text{-representation}\}.$$

Example 4.3. The left regular representation is a $C_c(\Gamma)$-representation with the dense subspace $C_c(\Gamma) \subset \ell^2(\Gamma)$.

Since $C_c(\Gamma) \subset D$, the norm is well-defined. These D-representation have some properties that are stated below. The cases $D = \ell^p(\Gamma) \lhd \ell^\infty(\Gamma)$, $p \in [1, \infty]$ are of interest and studied in Section 3.

Definition 4.4. The completion of $\mathbb{C}[\Gamma]$ in the D-norm gives us a C*-algebra, $\mathrm{C}^*_D(\Gamma)$. For $D = \ell^p(\Gamma)$, we write $\mathrm{C}^*_{\ell^p}(\Gamma)$.

4.2 Properties of Intermediate Group Algebras

Now we will state a few properties of these intermediate group algebras $\mathrm{C}^*_{\ell^p}(\Gamma)$, and see how they behave under some basic operations such as the direct sums and tensor products.

Lemma 4.5 (Universal property of $\mathrm{C}^*(\Gamma)$)**.** There exists a surjection from the universal C*-algebra onto $\mathrm{C}^*_D(\Gamma)$ extending the identity map: $\mathrm{id} : \mathbb{C}[\Gamma] \to \mathbb{C}[\Gamma]$.

Lemma 4.6. If D is a closed ideal of $\ell^\infty(\Gamma)$, and π is a D-representation, then every matrix coefficient of that π belongs to D, i.e., $\pi_{\xi,\eta} \in D$ for all $\xi, \eta \in \mathcal{H}$, not just for elements in the dense subspace $\mathcal{H}_0$.

An example of a closed ideal of of $\ell^\infty(\Gamma)$ is $C_0(\Gamma)$.

Proof. We have that $\pi_{\xi,\eta} \in D$ *for all* $\xi, \eta \in \mathcal{H}_0$. Let $\alpha, \beta \in \mathcal{H}$, then there exists sequences of elements $\{\alpha_i\}$ and $\{\beta_i\}$ in $\mathcal{H}_0$ such that $\alpha_i \to \alpha$ and $\beta_i \to \beta$. Since D is closed, and the sequence of matrix coefficients in D converge to $\pi_{\alpha,\beta}$, $\{\pi_{\alpha_i,\beta_i}\} \to \pi_{\alpha,\beta}$ (sup norm), we get $\pi_{\alpha,\beta} \in D$. □

Lemma 4.7. The tensor product of a D-representation and an arbitrary representation is again a D-representation.

Proof. Suppose $\pi : \Gamma \to \mathbb{B}(\mathcal{H})$ is a D-representation and $\sigma : \Gamma \to \mathbb{B}(\mathcal{K})$ is arbitrary. For $v_i \in \mathcal{H}_0$ and $w_i \in \mathcal{K}$, we have

$$(\pi \otimes \sigma)_{v_1 \otimes w_1, v_2 \otimes w_2} = \pi_{v_1,v_2} \sigma_{w_1,w_2} \in D$$

since D is an ideal; further, such simple tensors have dense span in $\mathcal{H} \otimes \mathcal{K}$. □

Lemma 4.8. An arbitrary direct sum of D-representations is again a D-representation.

Proof. Let

$$\{\pi_i\} : \Gamma \longrightarrow \mathbb{B}(\mathcal{H}_i)$$

be D- representations with $\mathcal{H}_0^i$ as the respective dense subspace of $\mathcal{H}_0$. Then the direct sum

$$\sum_i \oplus \pi_i : \Gamma \longrightarrow \mathbb{B}(\oplus \mathcal{H}_i)$$

is a D-representation with the dense subspace $\oplus \mathcal{H}_0^i$ since D respects sums. □

For each $x \in \mathrm{C}_D^*(\Gamma)$, there exists a D-representation π such that $\pi(x) = 0$, since otherwise $\|x\|_D = 0$. Call this representation π_x and let $\pi = \sum \oplus \pi_x$. This gives us the existence of a faithful D-representation of $\mathrm{C}_D^*(\Gamma)$.

Definition 4.9. An ideal $D \lhd \ell^\infty(\Gamma)$ is called *translation-invariant* if it is invariant under both the left and right translation actions of Γ on $\ell^\infty(\Gamma)$.

Lemma 4.10. Every nonzero, translation-invariant ideal in $\ell^\infty(\Gamma)$ contains the ideal $c_c(\Gamma)$.

Proof. Let $D \lhd \ell^\infty(\Gamma)$ be any translation-invariant ideal. Let $f \in D$ be a non-zero element and let

$$\delta_m : \Gamma \longrightarrow \mathbb{C}$$

$$\delta_m(x) = 1 \text{ if } x = m, \text{ and } 0 \text{ otherwise.}$$

Now, since D is an ideal $(f^{-1}\delta_m)f = \delta_m$ for at least one m where $f(m) \neq 0$. Thus $\delta_m \in D$. D is translation-invariant and δ_ms form a basis for $c_c(\Gamma)$, hence we have $c_c(\Gamma) \subseteq D$. □

Lemma 4.11. Let D be a translation invariant ideal. If v is a cyclic vector for a representation and $\pi_{v,v} \in D$, then π is a D-representation.

Proof. We have $[\pi(\Gamma)v] = \mathcal{H}$. Let $\mathcal{H}_0 = span\{\pi(s)(v) : s \in \Gamma\}$, (a dense subspace), $\xi = \pi(g_1)(v)$ and $\eta = \pi(g_2)(v)$ in $\mathcal{H}_0$. Then, we have

$$\pi_{\xi,\eta}(s) = \langle \pi(s)\xi, \eta\rangle = \langle \pi(sg_1)\xi, \pi(g_2)\eta\rangle = \pi_{v,v}(g_2^{-1}sg_1)$$

so that $\pi_{\xi,\eta} \in D$. □

Lemma 4.12. Let D be a translation invariant ideal and $h \in D$ be a positive definite function. Then, the GNS representation corresponding to h is a D-representation.

Proof. Let $h : \Gamma \to \mathbb{C}$ be a positive-definite function and let $\pi : \Gamma \to \mathcal{U}(\mathcal{H}_\langle)$ be the GNS associated to h with the corresponding cyclic vector v. We have $\pi_{v,v} = h$ and now, using the previous lemma we have that $\pi_{v,v} \in D$, i.e., π is a D-representation. □

Proposition 4.13. Let $\phi : \Gamma_1 \to \Gamma_2$ be a group homomorphism and let $D_i \lhd \ell^\infty(\Gamma_i)$ be ideals satisfying the following condition: **if** $f \in D_2$ **then** $f \circ \phi \in D_1$**.** Then ϕ extends to a C*- homomorphism $\mathrm{C}^*_{D_1}(\Gamma_1) \to \mathrm{C}^*_{D_2}(\Gamma_2)$.

Proof. We have the following diagram

$$\begin{array}{ccc} \Gamma_1 & \xrightarrow{\phi} & \Gamma_2 \\ & \searrow^{\pi\circ\phi}_{D_2 rep} & \downarrow{\scriptstyle \pi}_{D_1 rep} \\ & & \mathbb{B}(\mathcal{H}) \end{array}$$

where ϕ is group homomorphism, and for any $f \in D_2$, we have $f \circ \phi \in D_1$. The map $\pi \circ \phi$ extends to

$$\pi \circ \phi : \mathrm{C}^*_{D_1}(\Gamma) \to \mathbb{B}(\mathcal{H})$$

with

$$(\pi \circ \phi)_{\xi,\eta}(s) = \pi_{\xi,\eta} \circ \phi(s).$$

Thus, for any D_1-representation π, we have $\pi \circ \phi$ to be a D_2-representation. Thus we can extend ϕ to a C*- homomorphism

$$\phi : \mathrm{C}^*_{D_1}(\Gamma_1) \to \mathrm{C}^*_{D_2}(\Gamma_2).$$

□

Proposition 4.14. The following assertions hold:

1. Let $D_i \lhd \ell^\infty(\Gamma)$ be ideals such that $D_2 \subset D_1$. There exists a quotient map $\mathrm{C}^*_{D_1}(\Gamma) \to \mathrm{C}^*_{D_2}(\Gamma)$ (extending the identity map on $\mathbb{C}[\Gamma]$).

2. Suppose $\Lambda \subset \Gamma$ is a normal subgroup with $D_2 \lhd \ell^\infty(\Gamma/\Lambda)$ and $D_1 \lhd \ell^\infty(\Gamma)$. If $Im(D_2) \subset D_1$ under the inclusion $\phi : \ell^\infty(\Gamma) \to \ell^\infty(\Gamma/\Lambda)$, then there exists a surjection $\mathrm{C}^*_{D_1}(\Gamma/\Lambda) \to \mathrm{C}^*_{D_2}(\Gamma)$ extending ϕ.

Proof. 1. Let $\phi = \mathrm{id}$ be the identity map on Γ. Using the above 4.13, we have that for any $f \in D_2$, $f \circ \mathrm{id} \in D_2 \subset D_1$. Thus, we get that the identity map extends to a C*- homomorphism $\mathrm{C}^*_{D_1}(\Gamma) \to \mathrm{C}^*_{D_2}(\Gamma)$.

2. Let ϕ be the quotient map $\Gamma) \to \Gamma/\Lambda$. Again, we apply the previous proposition 4.13. If $f \in D_2 \subset \ell^\infty(\Gamma)$ then $\Gamma \xrightarrow{\phi} \Gamma/\Lambda \xrightarrow{f} \mathbb{C}$ and $f \circ \phi \in D_2$, hence ϕ extends to a C*- homomorphism $\mathrm{C}^*_{D_1}(\Gamma/\Lambda) \to \mathrm{C}^*_{D_2}(\Gamma)$.

□

5. PROPERTIES OF $\mathrm{C}^*_{\ell p}(\Gamma)$

5.1 $\mathrm{C}^*_{\ell p}(\Gamma)$, $\mathrm{C}^*_r(\Gamma)$and $\mathrm{C}^*(\Gamma)$

We start with a very basic and direct consequence of the construction.

Lemma 5.1. For every discrete group Γ, the completion with respect to the ideal $\ell^\infty(\Gamma)$ is the universal (or full) group C*-algebra: $\mathrm{C}^*_{\ell^\infty(\Gamma)}(\Gamma) = \mathrm{C}^*(\Gamma)$.

Proof. Since every matrix coefficient associated to a unitary representation of Γ is an $\ell^\infty(\Gamma)$ representation, the universal and $\ell^\infty(\Gamma)$ norms coincide.

$$\|x\|_{\ell^\infty(\Gamma)} := \sup_\pi \{\|\pi(x)\|_{\mathbb{B}(\mathcal{H})} : \pi \text{ is a } \ell^\infty(\Gamma)\text{-representation}\}$$

$$= \sup_\pi \{\|\pi(x)\|_{\mathbb{B}(\mathcal{H})} : \pi \text{ is a unitary representation}\} = \|x\|_{uni}.$$

□

Theorem 5.2. *[2, Proposition 2.11] For every discrete group Γ and every $p \in [1, 2]$ the completion with respect to the ideal $\ell^p(\Gamma)$ is the reduced group* C*-algebra*: $\mathrm{C}^*_{\ell^p}(\Gamma) = \mathrm{C}^*_r(\Gamma)$.

The proof of this requires some prerequisites first and a result by Cowling, Haagerup and Howe from ([4, Theorem 1]). This theorem is proven for a

locally compact group. The proof for the special (and simpler) case of a discrete group is given below.

Lemma 5.3 (Weak containment of representations)**.** Let σ and π be two unitary representations

$$\sigma : \Gamma \longrightarrow \mathbb{B}(\mathcal{H})$$

$$\pi : \Gamma \longrightarrow \mathbb{B}(\mathcal{K}).$$

Let ker π and ker σ denote the kernels of their respective representations of $C^*(\Gamma)$. Then we say that σ is weakly contained in π, denoted $\sigma \prec \pi$ if any of the following conditions hold true:

1. For each $\xi \in \mathcal{H}$, $F \subset \Gamma$ finite, and $\epsilon > 0$, there exists $\eta_1, ..., \eta_n \in \mathcal{K}$ such that
$$|\langle \sigma(\gamma)\xi, \xi\rangle - \sum_{j=1}^{n} \langle \pi(\gamma)\eta_j, \eta_j\rangle| < \epsilon$$
for all $\gamma \in F$.

2. $\|\sigma(f)\| \leq \|\pi(f)\|$ for all $f \in \ell^1(\Gamma)$. (It is enough to show it for all $f \in C_c(\Gamma)$ by density.)

3. ker $\pi \subset$ ker σ.

Proposition 5.4. Let σ be a unitary representation of the discrete group Γ on the Hilbert space $\mathcal{H}_\sigma$.

$$\sigma : \Gamma \longrightarrow \mathcal{U}(\mathcal{H}_\sigma)$$

Let ξ be a cyclic vector in $\mathcal{H}_\sigma$ (i.e., the set of vectors $\{\sigma(s)\xi : s \in \Gamma\}$ spans a dense subspace of $\mathcal{H}_\sigma$). Suppose that the diagonal matrix coefficient

$\phi : x \to \langle \sigma(x)\xi, \xi \rangle$ lies in $\ell^{2+\varepsilon}(\Gamma)$ for each ε in $\mathbb{R}^+$. Then σ is weakly contained in the regular representation λ.

Proof. We will prove that

$$||\sigma(f)|| \leq ||\lambda(f)||, \quad \text{for all } f \in C_c(\Gamma).$$

Let ρ be a unitary representation $\rho : \Gamma \to \mathcal{U}(\mathcal{H}_\rho)$. Since $||\rho(f^*f)|| = ||\rho(f)||^2 = ||\rho(f)^*\rho(f)||$ for all $f \in C_c(\Gamma)$, we work only with $f \in C_c(\Gamma)$ such that $f = f^*$. Since f is self adjoint, we have $\rho(f) = \rho(f)^*$. Since $\rho(f)$ is a self adjoint operator on $\mathcal{H}_\rho$, there exists a unique spectral measure for $\theta \in \mathcal{H}_\rho$, of the form

$$\mu_{\theta,\theta} : \mathcal{B}(\sigma(\rho(f))) \to \mathcal{P}$$

, where $\mathcal{B}(\sigma(u))$ denotes the Borel sets of $\sigma(\rho(f))$ and $\mathcal{P}$ denote the projections in $\mathbb{B}(\mathcal{H}_\rho)$. $\mu_{\theta,\theta}$ satisfies $\int \rho(f) d\mu_{\theta,\theta} = \langle \rho(f)\theta, \theta \rangle$.

By Holder's inequality, we have

$$\int t^2 \mu_{\theta,\theta} \leq \left(\int t^{2n} d\mu_{\theta,\theta} \right)^{1/n} \left(\int d\mu_{\theta,\theta} \right)^{1-1/n}.$$

Moreover, we see that

$$\lim_{n\to\infty} \left(\int t^n d\mu_{\theta,\theta} \right)^{1/n} = \sup\{t^2 \mid t \in \text{supp}(\mu_{\theta,\theta})\}.$$

and

$$||\rho(f)\theta|| = \langle \rho(f^*)^2\theta, \theta \rangle^{1/2} \leq \lim_{n\to\infty} \langle \rho(f^{(*2n)})\theta, \theta \rangle^{1/2n} ||\theta||.$$

For a dense subspace H of H_σ,

$$\|\rho(f)\| = \sup_{\theta\in H} \|\rho(f)\theta\|\|\theta\|^{-1} \leq \sup_{\theta\in H} \lim_{n\to\infty} \langle \rho(f^{(*2n)})\theta, \theta\rangle^{1/2n} \leq \|\rho(f)\|.$$

f is of the form $f = \sum_{s\in\Gamma} \alpha_s s$ and $f^* = \sum_{s\in\Gamma} \overline{\alpha}_{s^{-1}} s$.

$$f^{*(2n)} = (f * f^*)^n, \quad f * f^* = \sum_{s,t\in\Gamma} \alpha_s \overline{\alpha}_{t^{-1}} st$$

$$\rho(f * f^*) = \sum_{s,t\in\Gamma} \alpha_s \beta_t \rho(s).$$

$$||\rho(f)|| = \sup_{\theta\in H} \lim_{n\to\infty} \sum_{s\in\Gamma} (f * f^*)^{(*2n)}(s)((\rho(s)\theta, \theta))^{1/4n}. \tag{5.1}$$

We have an expression for $||\rho(f)||$ for a unitary representation ρ and $f \in C_c(\Gamma)$. Now, we use this for σ and λ.

1. For $\rho = \sigma$

   $\mathcal{H} := \{\sigma(s)\zeta \mid s \in C_c(\Gamma)\}$, a dense subspace of H_σ. Fix $g \in C_c(\Gamma)$ since ζ is a cyclic vector. Let

   $$g = \sum_{s\in\Gamma} \alpha_s s \in C_c(\Gamma), \quad \overline{\sum_{s\in\Gamma} \alpha_s s} := \sum_{s\in\Gamma} \overline{\alpha_s} s^{-1}.$$
   $$\theta = \sigma(g)\zeta = \sum_s \alpha_s \sigma(s)\zeta,$$

   θ is an element of the dense subspace $\mathcal{H}$. For all $s \in \Gamma$, define

   $$\phi(s) := \langle \sigma(s)\zeta, \zeta\rangle,$$
   $$\psi(s) := \langle \sigma(s)\theta, \theta\rangle.$$

We have

$$\begin{aligned}\psi(s) &:= \langle \sigma(s)\theta, \theta\rangle \\ &= \sum_s \alpha_s \overline{\alpha}_s \langle \sigma(s)\sigma(s)\zeta, \sigma(s)\zeta\rangle \\ &= \sum_s |\alpha_s|^2 \langle ((\sigma(s))^*\sigma(s)\sigma(s)\zeta, \zeta\rangle. \\ &= \sum_s |\alpha_s|^2 \langle \sigma(s^{-1})\sigma(s)\sigma(s)\zeta, \zeta\rangle. \\ &= \sum_s |\alpha_s|^2 \langle (\sigma(1)\sigma(s)\zeta, \zeta\rangle. \\ &= \sum_s |\alpha_s|^2 \langle \sigma(s)\zeta, \zeta\rangle.\end{aligned}$$

Since $\langle \sigma(s)\zeta, \zeta\rangle \in \ell^{2+\varepsilon}(\Gamma)$ and since the summand here consists of finite terms, for any $\varepsilon > 0$, the map

$$\psi : s \mapsto \langle \sigma(s)\theta, \theta\rangle \in \ell^{2+\varepsilon}(\Gamma).$$

Now, if $X \subseteq \Gamma$ is a finite subset such that $(f^* * f * f^* * f)(x) \neq 0$ for all $x \in X$, then we have for any $\epsilon > 0$,

$$\begin{aligned}|\sum_s (f^* * f)^{*2n}(s)\psi(s)|^2 &= |\sum_{x\in X} (f^* * f * f^* * f)^{*n}(x)\psi(x)|^2 \\ &\leq |\sum_{x\in X} (f^* * f * f^* * f)^{*n}(x)|^2 \sum_{x\in X} |\psi(x)|^2 \\ &\leq \|(f^* * f * f^* * f)^{*n}\|_2^2 \|\psi\|_{2+\epsilon}^2 |X^n|^{\frac{\epsilon}{8(1+\epsilon)}}.\end{aligned}$$

Using the inequality we got in 5.1:

$$||\rho(f)|| = \sup_{\theta\in H} \lim_{n\to\infty} \sum_{s\in\Gamma} (f * f^*)^{(*2n)}(s)\langle\rho(s)\theta,\theta\rangle^{1/4n},$$

we have

$$\|\sigma(f)\| \leq \liminf_{n\to\infty} \|(f^**f*f^**f)^{*n}\|_2^2 \|\psi\|_{2+\epsilon}^2 |X^n|^{\frac{\epsilon}{8(1+\epsilon)}} \leq \liminf_{n\to\infty} \|(f*f^*)^{*2n}\|_2^{\frac{1}{4n}}$$

2. For $\rho = \lambda$, the left-regular representation, we have our dense subspace $\mathcal{H}$ to be $C_c(\Gamma)$ in $\mathcal{H}_\lambda = \ell^2(\Gamma)$. For $g \in C_c(\Gamma)$,

$$|\sum_s (f^* * f)^{*2n}(s)\langle\lambda(s)g,g\rangle|^2 = \|(f * f^*)^{*2n}\|_2^2 \|\langle\lambda(s)g,g\rangle\|_2^2$$

Thus, from the above inequality 5.1 for ρ, we have

$$\|\lambda(f)\| = \sup_{g\in C_c(\Gamma)} \lim_{n\to\infty} \sum_{s\in\Gamma} (f*f^*)^{(*2n)}(s)\langle\lambda(s)\theta,\theta\rangle^{1/4n} \leq \liminf_{n\to\infty} \|(f*f^*)^{*2n}\|_2^{\frac{1}{4n}}$$

Also, by definition,

$$(f^* * f)^{(*2n)} = (f^* * f)^{(*2n-2)} * (f^* * f)^{(*2)}$$

and, so

$$\left\|(f^* * f)^{(*2n)}\right\|_2 \leqq \|\lambda\,(f^* * f)\|^{2n-2} \left\|(f^* * f)^{(*2)}\right\|_2,$$
$$\limsup_{n\to\infty} \left\|(f^* * f)^{(*2n)}\right\|_2^{\frac{1}{4n}} \leqq \|\lambda(f)\|.$$

Hence we can conclude that

$$\lim_{n\to\infty} \left\| (f^* * f)^{(*2n)} \right\|_2^{\frac{1}{4n}} = \|\lambda(f)\|$$

Comparing the equalities obtained for σ and λ, we have proved that

$$||\sigma(f)|| \leq ||\lambda(f)||, \quad \text{for all } f \in C_c(\Gamma).$$

□

Proof of Theorem 5.2. We have $\pi : \Gamma \longrightarrow \mathbb{B}(\mathcal{H})$ is weakly contained in the regular representation. For any x, we can find a cyclic representation π such that $\pi(x) \neq 0$. Since, π is weakly contained in λ, $C^*(\Gamma) \to C^*_{\ell^p}(\Gamma)$ has a trivial kernel. This gives us that

$$\|x\|_{\ell^p(\Gamma)} = \|x\|_r \quad \text{for} \quad p \in [1, 2].$$

□

5.2 A Non-trivial Example: $\Gamma = \mathbb{F}_n$

Now, to show that this construction is non-trivial, we give an example where the algebras $C^*_r(\Gamma), C^*(\Gamma)$ and $C^*_{\ell^p}(\Gamma)$ are all distinct.

Proposition 5.5. Let $\mathbb{F}_n$ be a free group with $n \geq 2$. There exists a $p \in (2, \infty)$ such that $C^*(\mathbb{F}_n) \neq C^*_r(\mathbb{F}_n) \neq C^*_{\ell^p}(\mathbb{F}_n)$

For the proof we state a few prerequisites. We start by understanding the

free group. Let $\mathbb{F}_n$ be the free group generated by $a_1, \cdots, a_n$. Any element $s \in \mathbb{F}_n$ is a finite expression of a_is and a_j^{-1}s where $i, j \in \{1, 2, \cdots, n\}$.

Denote by $|s|$ the length of s which is the number of a_is and a_i^{-j}s in the reduced expression of s. For instance, let $s = a_1 a_3^{-1} a_4 a_6 a_1 a_2^{-1}$ then $|s| = 6$. For $s, t \in \mathbb{F}_n,\ d(s,t) := |t^{-1}s|$. Clearly, $|s^{-1}| = |s|$ and $|st| \leq |s| + |t|$.

Now, we define negative definite functions and state equivalent characterizations from [1].

Definition 5.6. A function

$$\psi : \Gamma \longrightarrow \mathbb{C}$$

is negative-definite if for all $n \in \mathbb{N}$, and all n-tuples $\gamma_1, \cdots, \gamma_n \in \Gamma$, the matrix $[\psi(\gamma_i) + \overline{\psi(\gamma_j)} - \psi(\gamma_i \gamma_j^{-1})]$ is positive hermitian, i.e., for any n-tuple $(c_1, \cdots, c_n) \in \mathbb{C}^n$,

$$\sum_{i,j=1}^{n} (\psi(\gamma_i) + \overline{\psi(\gamma_j)} - \psi(\gamma_i \gamma_j^{-1})) c_i \overline{c_j} \geq 0$$

Lemma 5.7. A function $\psi : \Gamma \to \mathbb{C}$ is negative definite if and only if the following three conditions are satisfied:

1. $\psi(0) \geq 0$
2. $\psi = \tilde{\psi}$, where $\tilde{\psi} = \overline{\psi(x^{-1})}$
3. For all $n \in \mathbb{N}$, $\gamma_1, \cdots, \gamma_n$ and $c_1, \cdots, c_n$,

$$\sum_{i=1}^{n} c_i = 0 \text{ implies } \sum_{i,,j=1}^{n} \psi(\gamma_i \gamma_j^{-1}) c_i \tilde{c_j} \leq 0$$

Theorem 5.8 (Schoenberg)**.** *A function $\psi : \Gamma \to \mathbb{C}$ is negative definite if and only if the following two conditions are satisfied:*

1. $\psi(0) \geq 0$
2. *The function $\gamma \mapsto \exp(-t\psi(\gamma))$ is positive definite for all $t > O$.*

The following lemma plays a crucial role in proving the main result in this section.

Lemma 5.9. [5, Lemma 2] Let $\lambda > 0$. The function

$$\mathbb{F}_n \longrightarrow \mathbb{R}$$

$$s \mapsto \exp(-\lambda|s|)$$

is a positive definite function on $\Gamma = \mathbb{F}_n$.

Proof. Let a_is be the generators in $\mathbb{F}_n$. Using the previous two results, we only need to show that, for $s \in \mathbb{F}_n$ the map $s \mapsto |s|$ is negative definite, i.e., given any $n \in \mathbb{N}$, $s_1, \cdots, s_n \in \mathbb{F}_n$ and scalars $c_1, \cdots, c_n$ with $\sum_{i=1}^n c_i = 0$ we have $\sum_{i,,j=1}^n |(s_i^{-1}s_j)|c_i\overline{c_j} \leq 0$. Define

$$\Lambda = \{(s,t) \in \mathbb{F}_n \times \mathbb{F}_n | s^{-1}t = a_i \text{ for some } i\}$$

$$\bar{\Lambda} := \{(s,t)|(t,s) \in \Lambda\}.$$

Note $\Lambda \cap \bar{\Lambda} = \phi$. Let $\mathcal{H}_\Lambda$ be a Hilbert space with orthonormal basis $(e_{(s,t)})_{(s,t)\in\Lambda}$. For $(s,t) \in \bar{\Lambda}$ put $e_{(s,t)} = -e_{(t,s)}$.

Now, let $r \in \mathbb{F}_n$ be an element with the word $r = r_1 r_2 \cdots r_m$ where each r_i is either a generator a_k or its inverse. This gives us that $|r| = m$. Now consider the kth truncations of r given by $s_k := r_1 r_2 \cdots r_k$ and $s_0 := e$. Note that $s_{k-1}^{-1} s_k = r_k$ is either some a_i or a_i^{-1} and so

$$(s_{k-1}^{-1}, s_k) \in \Lambda \cup \bar{\Lambda}.$$

Now we define a function

$$f : \mathbb{F}_n \longrightarrow \mathcal{H}_\Lambda$$

$$r \mapsto e_{s_0,s_1} + e_{s_1,s_2} + \cdots + e_{s_{m-1},s_m}$$

which is $f(r) = e_{s_0,s_1} + e_{s_1,s_2} + \cdots + e_{s_{m-1},s_m} = e_{e,r_1} + e_{r_1,r_1r_2} + \cdots + e_{r_1r_2\cdots r_{m-1},r_1r_2\cdots r_{m-1}r_m}$

Note that $e_{s_k,s_{k+1}}$ are orthogonal since each truncation of r is distinct and hence $\|f(r)\|^2 = m = |r|$.

Now, for $q, r \in \mathbb{F}_n$,

$$q = q_1 q_2 \cdots q_k,\ |q| = k \text{ and } r = r_1 r_2 \cdots r_m,\ |r| = m,$$

we have $q^{-1}r = q_k^{-1} \cdots q_2^{-1} q_1^{-1} r_1 r_2 \cdots r_m$. Let $l > 0$ be the smallest number such that $q_l = r_l$ but $q_{l+1} \neq r_{l+1}$. For instance, $q = a_4 a_2^{-1} a_3$ and $r = a_4 a_1 a_2 a_4^{-1}$, then $l = 1$ and $q^{-1}r = a_3^{-1} a_2 \underline{a_4^{-1} a_4} a_1 a_2 a_4^{-1}$.

Note, $0 \leq l \leq \min\{k, m\}$ and

$$q^{-1}r = q_k^{-1} \cdots q_{l+2}^{-1}\ q_{l+1}^{-1}\ r_{l+1}\ r_{l+2} \cdots r_m.$$

Also, both

$$f(q) = e_{e,q_1} + e_{q_1,q_1q_2} + \cdots + e_{q_1q_2\cdots q_{k-1},q_1q_2\cdots q_{k-1}r_k}$$

$$f(r) = e_{e,r_1} + e_{r_1,r_1r_2} + \cdots + e_{r_1r_2\cdots r_{m-1},r_1r_2\cdots r_{m-1}r_m}$$

have the same first l terms. Using the fact that $e_{(s,t)} = -e_{(t,s)}$ for $(s,t) \in \Lambda \cup \bar{\Lambda}$ we have that

$$|q^{-1}r| = \|f(q) - f(r)\|^2 = m + k - 2l.$$

Lastly, we also have that all the unit basis vectors $e_{(s,t)}$ have real scalar products and so

$$(f(s)|f(t)) \in \mathbb{R}.$$

Now, we return to the assumption that given any $n \in \mathbb{N}$, $s_1, \cdots, s_n$ and $c_1, \cdots, c_n$ with $\sum_{i=1}^n c_i = 0$, we have

$$\sum_{i,,j=1}^{n} |(s_i^{-1}s_j)|c_i\overline{c_j} = \sum_{i,,j=1}^{n} \|f(s_i) - f(s_j)\|^2 c_i\overline{c_j}$$

$$= \sum_{i,,j=1}^{n} (\|f(s_i)\|^2 + \|f(s_j)\|^2 - 2(f(s_i)|f(s_j)))c_i\overline{c_j}$$

The first two terms are 0 since $\sum_{i=1}^n c_i = 0$ and so the remaining sum is $-2\|\sum_k c_k f(s_k)\|^2 \leq 0$ giving us that $s \mapsto |s|$ is negative definite and thus $s \mapsto e^{-\lambda s}$ is positive definite on $\mathbb{F}_n$. □

Proof of Proposition 5.5. Now, we show that $\mathrm{C}^*_r(\Gamma) \neq \mathrm{C}^*_{\ell p}(\Gamma)$ for some p. Let

$$h_n(s) := e^{-|s|/n}.$$

Let $\Gamma = \mathbb{F}_m$ Each h_n is positive definite and $h_n \to 1$ pointwise. Fix $n \in \mathbb{N}$ and choose $p_n \in \mathbb{N}$ such that $m \leq e^{p_n/n}$, so that $m \leq e^{-p_n/n} \leq 1$.

$$\sum_{s\in\Gamma}(e^{-|s|/n})^{p_n} = \sum_{k=1}^{\infty}(\sum_{|s|=k} e^{-kp_n/n}) \leq \sum_{k=1}^{\infty}(me^{-p_n/n})^k < \infty.$$

Consider the respective GNS for each h_n,

$$\pi_n : C^*_{l_{p^n}}(\Gamma) \longrightarrow \mathbb{B}(\mathcal{H}_n)$$

with corresponding canonical cyclic vector v_n. Since $h_n = \pi_{v_n,v_n} \to 1$ pointwise,

$$\|\pi_n(s)v_n - v_n\| \to 0, \quad \forall s \in \Gamma.$$

Thus, the direct sum $\oplus\pi_n$ weakly contains the trivial representation. If $C^*_{l_{p^n}}(\Gamma) = C^*_r(\Gamma)$ for all n, then the direct sum is defined on $C^*_r(\Gamma)$. This gives us a contradiction since the non-amenablity of $\mathbb{F}_m$ prevents weak containment of the trivial representation. (Theorem 3.19, 2)

Thus, there exists some p_n for which $C^*_{l_{p^n}}(\Gamma) \neq C^*_r(\Gamma)$ □

5.3 Amenability of Γ

We now see how this construction helps answer a fundamental property of the underlying discrete groups, namely amenability.

Proposition 5.10. If there exists $p \in [1, \infty)$ for which $C^*(\Gamma) = C^*_{\ell p}(\Gamma)$, then Γ is amenable.

For the proof we need a lemma which we state next:

Lemma 5.11 (Glimm's Lemma)**.** [3, Lemma 1.4.11] Let $\mathcal{A} \subset \mathbb{B}(\mathcal{H})$ be a separable C*-algebra which contains no non zero compact operators. If ϕ is a state on $\mathcal{A}$, then there exist orthonormal vectors $\{\xi_n\}$ such that

$$\langle a\xi_n, \xi_n \rangle \to \phi(a)$$

Proof of Proposition 5.10. If $C^*(\Gamma) = C^*_{\ell^p}(\Gamma)$, then there exists a faithful ℓ^p representation of $C^*(\Gamma)$, say π. We can assume that $\pi(C^*(\Gamma))$ has no non-zero compact operators (by taking $\mathcal{H} \otimes \ell^2(\Gamma)$ instead of Γ if necessary).

Applying Glimm's lemma now, we have that

$$\langle (\pi(a) - a)\xi_n, \xi_n \rangle \to 0,$$

i.e., the trivial representation is weakly contained in π. Let v_n be unit vectors such that $\|\pi(s)v_n - v_n\| \to 0$ for all $s \in \Gamma$. Since π is a $\ell^p(\Gamma)$-representation, we can approximating v_n's with vectors having associated matrix coefficients in $\ell^p(\Gamma)$ and so we can assume $\pi_{v_n,v_n} \in \ell^p(\Gamma)$ for all n. Hence, we have positive definite functions $h_n = \pi_{v_n,v_n}$ in $\ell^p(\Gamma)$ tending to 1 pointwise. Using Theorem 3.19 (3), we need finitely supported positive definite functions tending to 1. We consider for $k > p$, the functions $h_n^k \in \ell^1(\Gamma) \subset C^*_r(\Gamma)$ with $h_n^k \to 1$ pointwise. Thus they can be approximated by finitely supported functions $f_n \in C_c(\Gamma)$ in the $\|\cdot\|_r$ norm. The square root of h_n^k can be approximated by $f_n * f_n^*$ and h_n^k is approximated by the finitely supported positive definite functions $f^* * f$ giving us that Γ is amenable. □

The next proposition helps us prove the converse to this giving us a

characterization of amenability in terms of the group C*-algebras.

Theorem 5.12. *Suppose $D \lhd \ell^\infty(\Gamma)$ is a translation invariant ideal. We have $C^*(\Gamma) = C^*_D(\Gamma)$ if and only if there exist positive definite functions $h_n \in D$ with $h_n \to 1$ pointwise.*

Proof. If $C^*(\Gamma) = C^*_D(\Gamma)$, then the proof of Proposition 5.10 works with the same construction of h_n's even in this case with $\ell^p(\Gamma) = D$.

Conversely, if there exist positive definite functions $h_n \in D$ with $h_n \to 1$ pointwise. We first show that vector states coming from D-representations are weak-$*$ dense in the state space of $C^*(\Gamma)$. Let $\phi : C^*(\Gamma) \to \mathbb{C}$ be a state. Define:

$$\phi_n : C^*(\Gamma) \longrightarrow \mathbb{C}$$

$$\phi_n(\sum_s \alpha_s s) := \sum_s \alpha_s h_n(s)\phi(s).$$

Note that $\phi_n = \phi \circ m_{h_n}$ where m_{h_n} is the Schur multiplier associated to h_n. Hence, ϕ_n defines a state on $C^*(\Gamma)$. Since D is an ideal and $h_n \in D$, we have $\phi_n|_\Gamma \in D$. The GNS associated to ϕ_n is therefore a D-representation. $\phi_n \to \phi$ (weak-$*$) as $\{\phi_n\}$s are uniformly bounded. Thus the vector states coming from D-representations $\{\pi_{\xi,\xi}\}$ with π being D-representations are weak-$*$ dense in the state space of $C^*(\Gamma)$. This implies that the kernel of

$$f : C^*(\Gamma) \longrightarrow C^*_D(\Gamma)$$

is trivial since if there exists $x \neq 0$ and $x > 0$ in $C^*(\Gamma)$ with $f(x) = 0$, we

have some $\sum_s \alpha_s s \in \mathbb{C}[\Gamma]$ such that

$$x = \lim_{uni} \sum_s \alpha_s s \neq 0 \quad f(x) = \lim_D \sum_s \alpha_s s$$

where the limits are in the universal and D-norm respectively.

$$x \neq 0 \implies \omega(\sum_s \alpha_s s) \neq 0.$$

There exists vectors states coming from D-representations $\{\pi^i_{\xi,\xi}\}_i$ with

$$\pi^i_{\xi,\xi}(\sum_s \alpha_s s) \longrightarrow \omega(\sum_s \alpha_s s) \ \ \text{(weak-}*\text{)}.$$

But, since

$$f(x) = 0 \implies \pi^i_{\xi,\xi}(\sum_s \alpha_s s) \to 0,$$

we get a contradiction giving us that $x = 0$ and ker $(f) = 0$.

Thus, $\mathrm{C}^*(\Gamma) = \mathrm{C}^*_D(\Gamma)$. □

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